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Executive Summary

Starting a semiconductor venture—from **architecture design to commercial chip production**—is a monumental, multi-disciplinary endeavour. It requires **robust design R&D**, proven IP (e.g. CPU cores, RISC-V, memory), advanced EDA toolchains, and **foundry partnerships** for fabrication and packaging. Foundry engagement typically proceeds via MPW (multi-project-wafer) prototyping ($5–100K, 3–6 month turnaround) or direct runs (NRE $0.5–20M, long lead-times)【46†L19-L23】【46†L81-L89】.

Capital needs are enormous: **advanced fabs** cost billions (e.g. a new 3nm fab ~$15–20B, with lithography tools *per tool* up to $380M【57†L25-L28】). Cleanrooms must meet *ISO 14644 Class 1–4* standards【26†L90-L98】, consuming vast ultra-pure water (millions of litres) and power. Over 200 specialty chemicals (HF, H_2SO_4, NMP, etc.) are used【59†L182-L190】【59†L249-L258】, requiring stringent waste treatment (HF alone generates ~40% of fab hazardous waste【59†L249-L258】). Skilled engineers across VLSI design, test, process engineering, and safety are essential.

Timeline: A rough schedule from concept to prototype is on the order of **1–2 years**【46†L19-L23】. Key phases include architecture/IP selection, RTL and DFT design, PDK integration, tape-out, and silicon bring-up. Packaging, test, yield optimization and regulatory approvals add further months. Scaling to volume production typically follows validation runs and qualification (as in foundry "first silicon" programs).

Costs and yield: Example costs (Silicon Analysts) show a 300 mm 28 nm wafer is ~$3K【46†L109-L118】, with mask sets $0.8–1.5M【46†L109-L118】. Expected yields vary (50–90% or more) depending on defect densities and redundancy. Packaging and test add (say) $1–5 per die. These factors combine into a **cost-per-good-die** formula that is highly sensitive to die area and yield. For instance, halving die area doubles dies-per-wafer and usually *raises yield*, thus roughly quartering cost-per-die【11†L335-L343】【46†L107-L115】.

Manufacturing model: Given the huge up-front investment, *fabless* is almost mandatory for a startup. This means outsourcing fabrication and packaging to foundries/OSATs. A fabless model channels capital into design/IP, with agility and lower fixed cost【43†L95-L104】【43†L112-L120】. Partnerships with EDA/IP vendors (Cadence, Synopsys, ARM, SiFive, etc.) and foundries (TSMC, GF, UMC, Samsung) are crucial. Government programs (like the US CHIPS Act, Eurofoundries) can offset costs via grants or shared prototyping (MPW/shuttle programs).

Recommendation: Begin as a fabless design house. Use proven open-source cores (RISC-V, Arm under flexible access), mature EDA flows, and join an MPW run for first silicon【7†L104-L113】【46†L33-L41】. Design for testability (DFT, MBIST) and include redundancy if feasible. Engage early with a foundry (via MPW or broker) to obtain the PDK and plan mask costs. Rigorous yield and cost modeling (Poisson/cluster yield models【13†L69-L77】【46†L107-L115】) should inform design trade-offs (e.g. SRAM redundancy vs area).

Above all, clearly document all assumptions (die size, defect density, mask/pricing quotes), track risks (supply chain, technical re-spins, regulatory) and set decision gates (e.g. R&D milestone, yield target, funding round) to guide the venture.

1. Chip Design Roadmap

1.1 Concept & Specification. Define chip architecture (MCU, DSP, AI accelerator, etc.) based on target market. Gather requirements (performance, power, area). Survey existing IP: CPU cores (Arm, RISC-V from SiFive or open cores), accelerators, memory (SRAM/Flash macros), analog blocks. A startup often leverages licensed IP to save time【7†L108-L117】.

1.2 R&D and Architecture. Perform system modeling and architectural trade-off analysis (e.g. roofline performance model). Plan features (cache sizes, DMA, peripherals). Sketch block diagrams. Evaluate process technology: smaller node = higher speed but drastically higher mask/wafer cost【46†L107-L115】 and advanced cleanroom needs. Typically 28–65nm is cost-effective for MCUs/AI-tiny designs.

1.3 RTL Design & Verification. Develop RTL (Verilog/VHDL) for CPUs, accelerators, system logic. Write testbenches and simulate at module and system level. Use formal methods or emulation for verification. Incorporate IP: integrate processor cores, libraries. Pay attention to design-for-test (DFT): insert scan chains, BIST for memories and logic to ensure post-silicon testability【39†L75-L83】. Plan MBIST for each SRAM macro. Verification must cover functionality, timing (setup/hold), power, and corner PVT conditions.

1.4 EDA Tools & PDK Setup. Obtain the Foundry PDK (Process Design Kit) under NDA. The PDK contains device models, design rules, SPICE models, etc. Early NDA/contact may take weeks【46†L19-L23】. Install EDA flows (synthesis, place&route, LVS/DRC checking, simulation) matching the PDK. Common vendors: Cadence, Synopsys, Siemens (Mentor). Open-source flows (OpenROAD) exist for some nodes, but commercial EDA is required for 5–28nm. Ensure access to power/performance simulators (SPICE, Voltus, Tempus).

1.5 RTL-to-GDSII Flow. Run logic synthesis to gate netlist (map to standard-cell library from PDK). Perform floorplanning (block placement, core area, I/O ring). Place & route into a GDSII layout. Check DRC (design-rule compliance) and LVS (layout versus schematic). Generate a GDSII or equivalent for mask tape-out. If using a VCA/intermediary, they may handle DRC flow.

1.6 DFT, Test and MBIST. Complete design-for-test: add scan chains to logic, MBIST controllers for each SRAM, and test access (JTAG). Simulate stuck-at tests, verify MBIST detects bitflips. Work with a design house (or EDA tools) to generate ATPG patterns. Plan boundary-scan for I/Os. Ensure the PDK cell libraries include test-characterization data.

1.7 Packaging & PCB Co-Design. Decide package type (e.g. QFN, BGA, WLCSP). Engage with an OSAT on mechanical drawing. Package affects pinout planning. Plan the printed-circuit board for prototypes. Early packaging consideration avoids late flips (e.g. if board needs tight decoupling or thermal pad). Estimate package/test cost: simple QFN (~\$0.10–1 per die at volume) vs complex multi-chip modules.

1.8 MPW vs Dedicated Run. For first silicon, use a multi-project wafer (MPW/shuttle) to minimize NRE【46†L33-L41】. MPWs run on fixed schedules and share masks: cost as low as \$5K–100K depending on die size/node【46†L33-L41】. This yields only a small batch (10–200) of dies【46†L33-L41】. If proof-of-concept is successful, plan a dedicated run for production (requires \$0.5–15M mask NRE【46†L33-L41】 and >1K wafer commitments【46†L70-L79】).

1.9 Yield Modeling. Throughout design, perform yield and cost modeling. The simple *Poisson yield* model Y=exp(-A·D) (die area A, defect density D) is a baseline【13†L63-L72】. More realistic is a *negative-*

binomial (clustered defects) or Murphy's model【13†L63-L72】. Use Monte Carlo spatial defect simulation to predict yield variations and wafer maps. Track how changes (e.g. area, redundancy) affect expected yield. For example, if die area is halved, Poisson yield roughly increases from exp(-A1D) to exp(-0.5A1D), improving outcome【11†L335-L343】.

1.10 Test, Qualification & Production. After receiving wafers, perform wafer probe (parametric test). Sort dies into bins (functioning, speed bins, etc). Package selected good dies and do final test. Characterize silicon: measure speed, power, yield. If yield or performance is poor, iterate design (expensive "respins"). Assuming pass, work with foundry/OSAT for volume ramp and qualification. For an ASIC, qualification to customer or standard (JEDEC, ISO) can take weeks.

2. Capital and Facilities

Building a chip **fab** is enormously capital-intensive. Even a "small" 200 mm fab costs on the order of \$1–3B, while an advanced 300 mm fab (7 nm/5 nm) can exceed \$10–20B. However, a startup almost always avoids fab ownership. Instead, budget for a clean **lab environment** and prototyping tools (small lithography, probe station, FIB, etc.) for validation.

- **Cleanroom:** Semiconductor processes require **ISO 14644 Class 1–4** air cleanliness【26†L90-L98】. This means <10–10^4 particles $\geq$0.1 µm/m^3. Class 3–6 areas cover most fab space (lithography, etch/dep), while assembly/test may use Class 5–7. Class rating dictates HEPA/ULPA filters, airflow, gowning, and continuous monitoring【26†L90-L98】【26†L114-L122】. Even minor contamination ruins wafers.

- **Equipment:** The **largest cost item** is lithography. For a cutting-edge node, an EUV scanner costs ~\$150–380M per tool【57†L25-L28】. DUV/immersion scanners (for 28nm and above) are \$50–100M each. A moderate node fab still needs i-line/KrF steppers (cheaper, \$1–5M each) for non-critical layers. Other major tools: etchers (plasma/RIE, \$5–20M each), deposition (CVD/ALD for films, \$5–10M), CMP polishers (~\$5M), diffusion/ion-implant (~\$5–10M), oxidation furnaces (\$2–5M). Metrology (CD-SEM, film-thickness measurement) and inspection tools are also multi-million-dollar items.

- **Utilities:** Fabs use vast resources. Ultra-pure water (UPW) usage is **enormous** – on the order of **millions of liters per day** for a 300 mm fab【59†L182-L190】. UPW is needed for rinsing after etch/strip. Power consumption is typically >100 MW for a large fab (driving HVAC, tools, wafer processing). Gas supply (nitrogen, argon, silane, etc.) and vacuum systems are critical. Backup generators and filtered air (air-handlers costing millions) are needed. Environmental systems must treat exhaust (HF neutralization, etc.) and wastewater (reverse osmosis, waste discharge treatment) to comply with local regulations.

- **Chemical & Waste Handling:** Over **200 chemicals** (solvents, acids) are used in lithography and etch【59†L182-L190】. Key reagents include hydrofluoric acid (HF), sulfuric acid, NMP, APTES, TMAH, and solvents. HF stands out: the Semiconductor Industry Association notes HF waste is ~40% of fab hazardous effluent【59†L249-L258】. Special SDS procedures, corrosion-resistant piping, and *waste neutralization plants* (on-site incineration or chemical treatment) are mandatory. Regulatory compliance is strict: U.S. EPA's Clean Water Act, OSHA standards for chemical hazards, and EU REACH in Europe would apply. Fabs often operate under regulatory permits detailing allowed discharges; violation risks fines.

- **Safety:** Training and PPE are critical. Workers must handle toxic/corrosive chemicals (HF is life-threatening), ultra-dry nitrogen (asphyxiation risk), and high-voltage equipment. Cleanroom gowning is rigorous. Safety systems (air showers, HF sensors, ventilation alarms) are required.

- **Footprint & Class:** In summary, a modern **fab requires multiple ISO-classified rooms** with contiguous support areas (chemical scrubbers, ventilation, equipment maintenance). Even a "pilot fab" (e.g. a few 300mm tools) might occupy 2000–5000 m^2. Full-scale fabs are 100,000+ m^2. The capital per square meter (or per tool) is very high.

3. Supply Chain & Vendor Ecosystem

A semiconductor startup relies on a **global ecosystem**:

- **IP Vendors:** Processors (ARM Ltd., or open RISC-V from SiFive, LowRISC), GPUs (Imagination, Vivante), DSPs (CEVA, Cadence Tensilica), and PHYs (Ethernet, DDR). Memory macros (SRAM, Flash) from Synopsys (Cortex-M series), Sanechips, or in-house. Security IP (crypto cores) from standards (e.g. CHERI). Each typically requires licensing fees (up-front & royalty).

- **EDA & Design Services:** Major EDA providers (Cadence, Synopsys, Siemens/EDA) supply tool suites for synthesis, PnR, verification, and analysis. Logic and timing libraries come with the PDK. Verification IP (for buses, DDR, USB, etc.) can be licensed. Turnkey design houses or consultants (e.g. Open-Silicon, Efabless) may be engaged if needed.

- **Foundries:** Pure-play foundries (TSMC, GlobalFoundries, UMC, SMIC, Tower, Vanguard) manufacture the chips. Selection depends on node, geography, capacity. Each foundry has its own process flavors, PDKs and qualification requirements. For example, 130nm–28nm are widely available; 14nm–5nm are restricted and very costly. Government export controls (e.g. US Export Administration Regulations) may restrict access to certain technologies (China/Taiwan tensions).

- **OSAT (Outsourced Semiconductor Assembly & Test):** Companies like ASE, Amkor, JCET, STATS ChipPAC handle packaging (wire-bond, flip-chip, advanced packaging like PoP or fan-out) and final test. They also do probing. A startup would contract with an OSAT for QFN or BGA packaging.

- **Materials and Equipment Suppliers:** Wafer vendors (SUMCO, Siltronic) for blanks; chemical suppliers (Dow, Merck, Versum) for specialty gases/liquids; photomask shops (Toppan, Photronics) for reticles. Tool OEMs (ASML for lithography, Lam Research/Kokusai/Tokyo Electron for etch/CVD, KLA/Tencor for inspection) may require service contracts.

- **Test & DFT Tools:** ATE companies (Advantest, Teradyne) provide chip-testing equipment. Yield analysis and statistical tools (Minitab, JMP) are useful. Test engineers may use pattern generators and oscilloscopes for lab debugging.

- **Standards Bodies:** Collaboration with standards and alliances (JEDEC for memory, USB-IF for USB, RISC-V Foundation, ISO) ensures compliance.

This ecosystem is interconnected: e.g. IP providers coordinate with foundries for PDK compatibility, and tool vendors keep pace with nodes in conjunction with foundries. The startup must manage NDAs with each party and secure licenses early.

4. Regulatory, Environmental and Workforce Considerations

Regulatory: Semiconductor fabs are heavily regulated. Clean air laws (e.g. US Clean Air Act) and water discharge permits (Clean Water Act) apply. Locally, authorities may require environmental impact assessments. Many semiconductor chemicals (HF, ammonia) are regulated under OSHA and REACH. Startup must budget for environmental engineers and compliance systems. Intellectual-property law is critical too: ensure freedom-to-operate for chosen IP and processes (patents in circuits, licenses for EDA/PDK must be carefully negotiated).

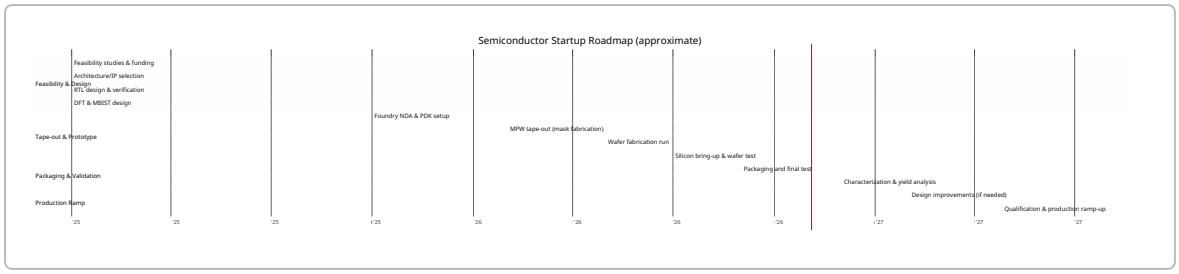
Environmental: Fabs consume huge water and energy; many jurisdictions (e.g. Singapore, South Korea) mandate closed-loop UPW recycling and energy recovery. Expect to install large wastewater treatment (chemical scrubbers, RO systems) and possibly green energy (onsite solar, or long-term power contracts). Monitoring of chemical emissions (volatile organic compounds, NO_x) is required by national regs.

Safety & Workforce: Operators need hazardous-materials (HAZMAT) training for acids, gases. Semiconductor work also demands cleanroom discipline. Technicians typically undergo months of training. A modern fab uses >1000 people (engineers, operators, maintenance). A lean startup fab (pilot line) might start with a few dozen, but larger production requires hundreds. Recruiting specialized talent (experienced VLSI designers, reliability/test engineers, process engineers) is difficult; often engineers come from existing semiconductor clusters or international relocations.

Infrastructure: The fab location must have reliable utilities (uninterrupted power, abundant water, clean internet connectivity) and be in a region with skilled talent pools (e.g. semiconductor hubs like Taiwan, Singapore, US's Silicon Valley, Europe's Dresden, Israel). Government incentives (tax breaks, grants) can be decisive: e.g. US CHIPS Act or EU's IPCEI programs.

5. Timelines and Milestones

A realistic project timeline from initial concept to first working silicon is **12–24 months**【46†L19-L23】. Key phases might look like:



Milestones: NDA/PDK secured; netlist sign-off; tape-out submission; silicon arrival; good-die yield meeting target; first packaged parts delivered.

All timelines have large uncertainties: fabrication queues at foundries can slip schedules, and yield issues can cause multi-month respins. Each phase should be a decision gate: if yield or performance is unacceptable at low volume, assess redesign vs changing specs.

6. Cost Models and Scenarios

We compare three scenarios: **MPW/Prototype**, **Small dedicated run**, and **Large-volume production**. Key cost components are masks (NRE), wafers, packaging/test, and overhead. Table below outlines typical values (2026):

Scenario	Process	Mask Cost	Wafer Cost	Die Area (assumed)	Yield (%)	Pack/Test Cost (per die)	Cost per *good* die (est.)
MPW Shuttle	28nm	\$0.8–1.5M	\$3,000	25 mm² (e.g.)	~50%	\$2/die	~\$100–200 (small n)
Low-volume run	65nm	\$0.3–0.5M	\$2,500	50 mm²	~70%	\$1/die	~\$50–100
High-volume prod	28nm	\$0.8–1.5M	\$3,000	25 mm²	~50%	\$1/die	~\$10–20 (excl. R&D amort.)

Sources: Wafer and mask costs from industry trackers【46†L109-L118】. Yields/packaging are illustrative. **Assumptions and unknowns:** Actual cost/die depends on design (area, complexity), volume, and negotiated pricing. For example, [46] estimates a 300 mm wafer at 28nm is \$3k and a mask set ~$0.8–1.5M【46†L109-L118】. Packaging/test (~\$1–3) is a small fraction of total per-die cost if volumes are high.

Notably, **die area** dominates: doubling area halves dies per wafer and (roughly) lowers yield, multiplying cost per die. For example, Poisson yield Y = exp(–A·D) means if area A doubles, yield nearly squares the loss. In practice, reducing die size gives more dies/wafer *and* higher yield【11†L335-L343】, dramatically cutting cost-per-good-die.

Sensitivity: Reducing process node (e.g. 65nm→28nm) increases mask and wafer cost by ~2× 【46†L109-L118】 but may justify higher performance. MPW vs dedicated: MPW amortizes mask cost across projects (lowers NRE burden) but yields few dies and no customization. Dedicated runs have huge NRE but lowest unit costs at scale.

Cost Breakdown: Roughly, for dedicated production, wafer fabrication is ~50% of cost per die, packaging+test ~10%, masks amortized <10% (at high volume), with the rest in overhead and profit margin. Exact split varies by node and yield.

7. Risks and Mitigation

- **Technical Risks:** Bugs or design flaws may cause silicon to fail. Mitigation: extensive verification and FPGA prototyping before tape-out. Manageably design complexity (modular IP reuse). Schedule design reviews (DFM review with foundry) to catch issues early.

- **Supply Risks:** Foundry capacity is constrained (AI demand). Secure foundry slots early; consider multiple fabs (e.g. TSMC and GF for 28nm). Inventory key inputs (e.g. dies, packaging, chip-grade substrates) to buffer short-term disruptions.

- **IP/Legal Risks:** Ensure all IP licenses are in place and clear. Use well-established IP cores to avoid re-spins. If using open RISC-V, check open-source licenses.

- **Regulatory/Environmental Risks:** Non-compliance can halt fab operations. Plan beyond minimum (install extra scrubbers, meet more stringent waste standards).

- **Market Risks:** Technology shifts (e.g. new standards or competitors) can erode product relevance. Mitigation: focus on niche/application or assemble strong partner network. Pre-sell or co-develop with an anchor customer if possible (e.g. early commitment from an electronics company).

- **Financial Risks:** Budget overruns due to schedule slips or re-spins can dilute funding. Maintain conservative burn-rate forecasts; tie funding tranches to milestones.

8. Organisation and Hiring

A startup chip company typically organises into:

- **Leadership/Engineering Management:** CEO/CTO (technology vision), CFO (finance/grants).
- **Architecture Team:** defines system specs.
- **RTL/Logic Design Team:** responsible for writing and simulating the design.
- **Physical Design Team:** does synthesis, PnR, DRC/LVS checks (often overlaps with CAD).
- **Test/Validation Team:** handles DFT, MBIST, chip bring-up and validation.
- **Verification Team:** functional and formal verification, EDA tool flow engineers.
- **Packaging/Test Team:** interfaces with OSATs, PCB designers.
- **Process/Modeling Team:** yield modeling, PDK and device modeling (may outsource).
- **Operations/Support:** manages IP contracts, legal, export compliance, procurement.
- **Sales/Marketing:** begin engaging potential customers (e.g. OEMs) early; gather feedback for specs.

Hiring must target experienced VLSI engineers. Often new startups recruit alumni from established semiconductor firms. Consider partnering with a university or incubator for design talent.

9. Funding Strategies

Options to finance the venture:

- **Venture Capital:** Many VCs are wary due to high costs and long cycles. Highlight unique IP or market demand (e.g. IoT/Mote markets, AI edge).
- **Strategic Partnerships:** Partner with a larger semiconductor/tech company for investment in exchange for product rights or co-development (e.g. automotive Tier-1s, AI startups).
- **Government Grants/Programs:** In US, the CHIPS Act provides subsidies for fabless initiatives and prototyping. EU offers similar grants (like Europractice for MPW access).
- **Foundry Programs:** Some foundries run accelerator programs (e.g. TSMC Open Innovation Platform) giving reduced access or support to startups.

- **IP "Design-First" Schemes:** Programs like Arm Flexible Access let startups license CPU cores with deferred payment【7†L104-L113】. Leverage these to minimize upfront licensing cash needs.
- **Commercial Revenue:** If applicable, early design services or licensing smaller IP blocks to generate cash flow.

Clearly list funding milestones: seed for prototyping, Series A for tape-out, etc. Allow for contingency funds (designs often need >1 tape-out for success).

10. Go-to-Market and Business Model

Most likely **fabless business model**: focus on design/IP; outsource manufacturing and assembly. This avoids CAPEX of a fab【43†L95-L104】. Alternative IDM model (building your own fab) is generally infeasible for a startup.

Key elements:

- **Value Proposition:** Define the chip's market (e.g. embedded microcontroller, AI accelerator). Emphasize differentiation (higher efficiency, integration, lower cost).
- **Customer Engagement:** Engage potential customers early (car manufacturers, IoT device makers, etc.). Use application-specific benchmarks or integration kits (FPGA prototypes) to demonstrate viability.
- **OSAT Relationships:** Select packaging/test vendors experienced in your target packaging. Negotiate volume discounts or shared risk agreements.
- **Standards & Qualification:** For certain markets (automotive, medical), chip qualification (AEC-Q100, ISO 26262 functional safety) is mandatory. Plan extra time/cost for qualification testing.
- **Revenue Models:** Primarily chip sales. Consider licensing core IP or design (if unique). Maintain IP control for IP licensing revenue.

Trade-off **fabless vs IDM**: Fabless minimizes fixed costs and maximizes flexibility【43†L95-L104】. An IDM approach (owning a fab) gives control but requires enormous capital and market volume (not advised for a new player). Similarly, chiplet or multi-chip module strategies can be explored as technology matures, but carry their own assembly/test complexity.

11. Case Studies and Precedents

- **SiFive (founded 2015):** A RISC-V startup that used VC funding and TSMC/GlobalFoundries foundries. They leveraged RISC-V's open ISA and ARM Ecosystem partnerships. Early tape-outs were on 130/28nm; they now serve markets like automotive AI accelerators. (No single citation, but industry accounts show SiFive raised >\$100M, announced chips by 2019.)

- **Hailo:** An AI chip startup (mentioned in [7]) that benefited from ARM's support. They focused on edge AI accelerators and emphasised proven IP to avoid re-spins【7†L114-L123】.

- **Arm Flexible Access:** Arm's program lets startups incorporate Cortex cores with deferred payments, substantially reducing up-front IP costs【7†L104-L113】. Many startups (e.g. Tenstorrent, Mythic) have used this to reach first silicon.

- **Small Foundry (AMIC, Tower):** Some fabless companies target older nodes (90–130nm) at smaller foundries (e.g. Tower or SkyWater) to reduce costs and complexity. For MCU products, this can be viable.

- **MPW Programs:** Universities and consortia (Europractice, MOSIS) have enabled startups and researchers to tape-out chips for \$10K–50K on 130nm–28nm. These programs are precedents for low-volume entry.

12. Deliverables & Decision Gates (Checklist)

1. **Market/Architecture Alignment**: Have clear target applications and customers.
2. **IP Agreements Secured**: CPU/accelerator licenses or open-source IP evaluated.
3. **PDK NDA Signed**: Foundry PDKs and design rules are in-house.
4. **RTL Complete & Verified**: Functional simulation coverage 100%; static timing met in simulation.
5. **DFT/ATPG Done**: Scan chains and MBIST included and passing.
6. **GDSII Sign-off**: LVS/DRC-clean layout ready for tape-out.
7. **Prototype Tape-out**: Submission to foundry (MPW or direct).
8. **Test Infrastructure Ready**: Probing/ATE at lab or via partner, board support.
9. **Silicon Yield Target Achieved**: Measured yield meets design assumptions.
10. **Characterization Done**: Meets all performance/power specs (PVT corners) in lab.
11. **Quality Review**: Documentation ready for customer (datasheets, reports).
12. **Packaging & Production Plan**: OSAT qualified, volumes planned, supply chain in place.
13. **Regulatory Compliance**: Environmental permits, safety standards, export controls satisfied.
14. **Funding Secured**: Financing aligned with production ramp needs.

Each gate should have “go/no-go” criteria. For example, if measured yield < target, re-evaluate need for design tweaks or process changes before proceeding to packaging.

Appendix: Key Contacts & Resources

- **Foundries:** TSMC, Samsung, GlobalFoundries, UMC (front-end); ASE, Amkor, JCET (back-end).
- **EDA/IP Vendors:** Cadence, Synopsys, Siemens, Arm, SiFive, CEVA.
- **OSATs:** ASE, Amkor, STATS (package/test).
- **Equipment:** ASML, TEL, Lam, KLA, Applied Materials.
- **Materials:** SUMCO, Siltronic (wafers); Merck, Dow (chemicals).
- **Standards Bodies:** JEDEC, RISC-V Foundation, SEMI, ISO.
- **RFP Checklist:** Ensure NDAs/PPA for PDK, mask specs, packaging specs, test vectors, yield criteria.

Assumptions and Unknowns: This report uses public data and industry estimates (e.g. Silicon Analysts pricing【46†L107-L115】) which may not reflect negotiated deals. Critical unknowns include actual SilicaFlux-V die area, target process node, and end-market volumes. All cost/yield figures are illustrative. If precise data (e.g. foundry quotes, defect densities) is lacking, sensitivity analysis should be run (e.g. yield = 30–90%, wafer cost ±20%).

Sources: Industry technical references and market reports have been used, including foundry engagement guides【46†L19-L23】【46†L107-L115】, semiconductor engineering analyses【43†L112-L120】, and standards documentation (ISO 14644【26†L90-L98】). Wherever possible, authoritative data from 2025–26 has been cited.